Name: _____

WORKSHEET

# Tangents to a curve

**1** Find the gradient of the tangent to the function $f(x) = x^2 - x + 6$ for the following values of $x$.

| $x$ | $-2$ | $-1$ | 0 | 1 | 2 |
|---|---|---|---|---|---|
| **Gradient of Tangent** | | | | | |

**2** Find the slope of the tangent to the curve $y = t^3 - 7t^2 + t - 2$ and complete the table.

| $t$ | $-2$ | $-1$ | 0 | 1 | 2 |
|---|---|---|---|---|---|
| **Slope of Tangent** | | | | | |

**3** Find where the gradient of $y = x^2 - 9x + 1$ s:

**a** $-1$

**b** 1

**c** 4

**d** $-11$

**4** Find where the gradient of $f(x) = x^3$ is equal to:

**a** 12

**b** 3

**c** 27

**d** 0

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**5** For what values of $x$ is the slope of $y = \frac{x^3}{3} - \frac{3}{2}x^2 - 6$ equal to:

**a** 10

**b** 4

**c** 0

**d** $-2$

**e** $-1$

**f** $-0.5$

**6** Find the angle (to nearest minute) that the tangent of $f(p) = 2p^2 - 7p - 5$ makes with the $x$-axis at:

**a** $p = 0$

**b** $p = 1$

**c** $p = 3$

**d** $p = \frac{7}{4}$

**7** Find the angle (to nearest minute) that the tangent of $f(x) = x^3 - 11x + 1$ makes with the $x$-axis at:

**a** $x = 0$

**b** $x = -1$

**c** $x = -2$

**d** $x = 1.95$

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**8** Find the equation of the tangent to $f(x) = x^3 - 3x^2 + 2x - 1$ at:

**a** $x = 0$

**b** $x = -1$

**c** $x = 1$

**d** $x = 3$

**9** Find the equation of the tangent to the curve $f(z) = -z^3 - 3z^2 + 10z + 1$ at:

**a** $z = -2$

**b** $z = 0$

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**c** $z = -0.5$

**d** $z = 1$

**10** Find the equation of the straight line perpendicular to the curve $f(x) = x^2 - 4x$ at:

**a** $x = -1$

**b** $x = 0$

**c** $x = 2$

**d** $x = 3$

## Answers

**1**

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| Gradient of Tangent | $-5$ | $-3$ | $-1$ | $1$ | $3$ |

**2**

| $t$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| Slope of Tangent | $41$ | $18$ | $1$ | $-10$ | $-15$ |

**3** **a** $x = 4$

**b** $x = 5$

**c** $x = \frac{13}{2}$

**d** $x = -1$

**4** **a** $x = \pm 2$

**b** $x = \pm 1$

**c** $x = \pm 3$

**d** $x = 0$

**5** **a** $x = 2, 5$

**b** $x = -1, 4$

**c** $x = 0, 3$

**d** $x = 1, 2$

**e** $x = \frac{3}{2} \pm \frac{\sqrt{5}}{2}$

**f** $x = \frac{3}{2} \pm \frac{\sqrt{7}}{2}$

**6** **a** $\theta = 98°8'$

**b** $\theta = 108°26'$

**c** $\theta = 78°41'$

**d** $\theta = 0°$

**7** **a** $\theta = 95°12'$

**b** $\theta = 97°8'$

**c** $\theta = 45°$

**d** $\theta = 22°10'$

**8** **a** $y = 2x - 1$

**b** $y = 11x + 4$

**c** $y = -x$

**d** $y = 11x - 28$

**9** **a** $y = 10z - 3$

**b** $y = 10z + 1$

**c** $y = 12.25z + 1.5$

**d** $y = 7z + 1$

**10** **a** $y = \frac{1}{6}x + \frac{31}{6}$

**b** $y = \frac{x}{4}$

**c** $x = 2$

**d** $y = -\frac{1}{2}x - \frac{3}{2}$

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